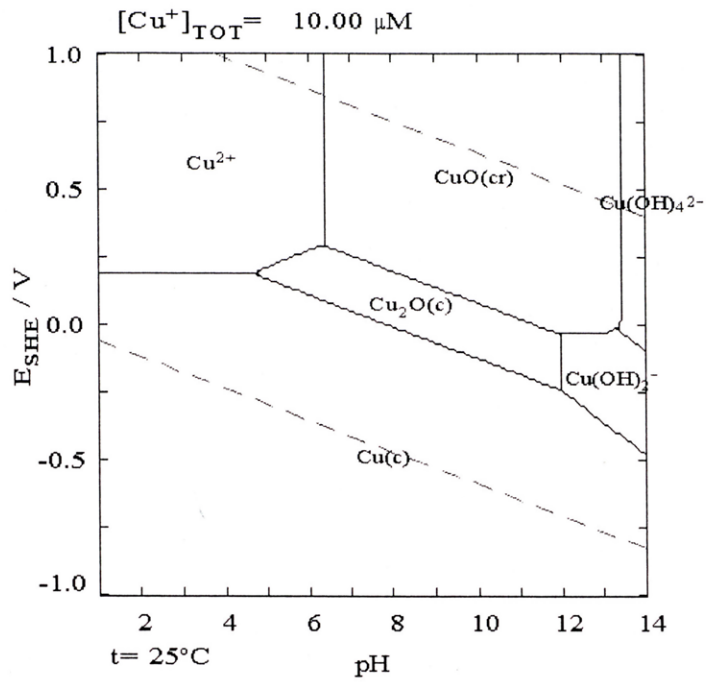
Inorganic Cumulative Exam

3-3-17

Answer five of the following questions. Only the first five of your answers will be graded.

Solid State and Materials Chemistry:

1) Calculate the lattice energy of the hypothetical compounds $NaCl_2$ and MgCl. You can assume that Na^{2+} and Mg^{+} ions have the same ionic radii as Na^{+} and Mg^{2+} respectively. Use the following data in your calculations. $Na^{+} \rightarrow Na^{2+}$ 4562 kJ/mol; $Mg^{+} \rightarrow Mg^{2+}$ 1451 kJ/mol; enthalpy of formation of NaCl, -411 kJ/mol; enthalpy of formation of $MgCl_2$, -642 kJ/mol.
2) Using the Pourbaix diagram for copper answer the questions below.

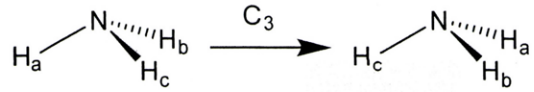


   a) Assuming a typical three electrode cell (working, counter and reference), what will happen to a copper working electrode in an aqueous 1M LiCl solution when the potential is swept from 0 V to +0.5 V vs SHE.
   b) What will happen to the same copper working electrode if this experiment is carried out in an aqueous 0.001 M LiOH solution
3) What materials and experimental conditions would you use to prepare colloidal Pt nanoparticles?
4) Describe the differences in bonding of any two allotropes of carbon.
5) Titania nanoparticles varying in size from 30-40 nm are currently used in the production of consumer sunscreen products. Describe at least two properties of these particles that would make them useful in this application.

Solution State Chemistry and Spectroscopy:

6) What is the point group of SF_4 ?

7) A partial transformation matrix for a C_3 rotation is given below. Provide a complete transformation matrix for the C_3 rotation of ammonia using the column vectors shown below.



$$\begin{bmatrix} & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & & & & 0 & 0 & 0 & & & \\ 0 & 0 & 0 & & & & 0 & 0 & 0 & & & \\ 0 & 0 & 0 & & & & 0 & 0 & 0 & & & \\ 0 & 0 & 0 & & & & & & & 0 & 0 & 0 \\ 0 & 0 & 0 & & & & & & & 0 & 0 & 0 \\ 0 & 0 & 0 & & & & & & & 0 & 0 & 0 \\ 0 & 0 & 0 & & & & & & & & & \\ 0 & 0 & 0 & & & & & & & & & \\ 0 & 0 & 0 & & & & & & & & & \end{bmatrix} \begin{bmatrix} X_O \\ Y_O \\ Z_O \\ X_{Ha} \\ Y_{Ha} \\ Z_{Ha} \\ X_{Hb} \\ Y_{Hb} \\ Z_{Hb} \\ X_{Hc} \\ Y_{Hc} \\ Z_{Hc} \end{bmatrix} = \begin{bmatrix} X'_O \\ Y'_O \\ Z'_O \\ X_{Hc} \\ Y_{Hc} \\ Z_{Hc} \\ X_{Ha} \\ Y_{Ha} \\ Z_{Ha} \\ X_{Hb} \\ Y_{Hb} \\ Z_{Hb} \end{bmatrix}$$

8) Provide the complete catalytic cycle for a Heck coupling reaction including all reactants and products.

9) Molybdenum hexacarbonyl has an A1 CO stretching frequency is 2001 cm^{-1} Tetrasodium tetracarbonylmolybdate has an A1 stretching frequency of 1463 cm^{-1} Why?

10) When cobalt(III) chloride hexahydrate is dissolved in ethanol, the solution is a deep blue color. When water is added to this deep blue ethanol solution, the color changes to a vibrant pink. Why did the color change? Be specific, include a description of the compounds involved and the electronic transitions. Be sure to discuss what orbitals are involved. (it may be helpful to start with an MO diagram of each compound).